Redox mechanisms: Worked solutions

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

E01

The left side has one oxygen atom; $\frac{1}{2}O_2$ on the right also has one. In the effective-charge bookkeeping, $0 = (+2) + 2(-1)$. Local neutrality under the stated assumptions gives $n = 2[V_O^{\bullet\bullet}] = 4 \times 10^{18}\,\mathrm{cm}^{-3}$. Other charged defects or dopants modify the balance, and a space-charge region need not be locally neutral; different vacancy charge states or carrier trapping also change the inference.

E02

The diffusion contribution is $-10^{-18} \times 10^{30} = -10^{12}\,\mathrm{m}^{-2}\mathrm{s}^{-1}$. The drift coefficient is $zeE/(k_BT) = (2)(1.60 \times 10^{-19})(10^8)/(4.14 \times 10^{-21}) \simeq 7.73 \times 10^9\,\mathrm{m}^{-1}$. Thus drift is $10^{-18}(10^{24})(7.73 \times 10^9) = +7.73 \times 10^{15}\,\mathrm{m}^{-2}\mathrm{s}^{-1}$. Adding diffusion gives $J \simeq +7.73 \times 10^{15}\,\mathrm{m}^{-2}\mathrm{s}^{-1}$ at the stated precision. Signs follow the stated positive x direction. Number flux is not itself electrical current density; $j_{\mathrm{ion}} = zeJ$.

E03

The uniform field magnitude is $0.6/12 = 0.05$ V/nm. Over 0.2 nm along the field, $\Delta\phi = -(0.05)(0.2) = -0.01$ V. For charge $+2e$, $\Delta U = ze\Delta\phi = -0.02$ eV. This does not specify the transition-state energy or location, so the activation-barrier change cannot be determined from endpoints alone.

E04

Divide the two expressions: $I_2/I_1 = e^{-2\kappa(g_2-g_1)} = e^{-2(5)(0.3)} = e^{-3} \simeq 0.0498$. The model holds the prefactor and κ fixed; changes in barrier shape or height with bias and composition, multiple transport paths, geometry and temperature can invalidate that assumption.

E05

$\tau_{\mathrm{th}} = R_{\mathrm{th}}C_{\mathrm{th}} = (10^6)(10^{-13}) = 10^{-7}$ s $= 100$ ns. The steady rise is $PR_{\mathrm{th}} = (10^{-4})(10^6) = 100$ K. At $t = \tau_{\mathrm{th}} = 100$ ns, $\Delta T = 100(1 - e^{-1}) \simeq 63.2$ K. A different pulse duration can deliver the same energy while heat leaves at a different rate; thermal geometry and boundary conditions also affect local peak temperature.

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Redox mechanisms: Worked solutions (continued)

E06

$E_a/k_BT = 1/0.02585 \simeq 38.68$. Hence $\tau = 10^{-12}e^{38.68} \simeq 6.3 \times 10^4$ s, or approximately 17.5 h. This is the timescale of the stipulated single process. Real retention also needs its actual parameters, operating temperature and definition of failure, distributions over cells and cycles, plus possible competing relaxation mechanisms.

E07

One oxidation is $\mathrm{Ag} \rightarrow \mathrm{Ag}^+ + e^-$; reduction reverses it, $\mathrm{Ag}^+ + e^- \rightarrow \mathrm{Ag}$. Electrons must be supplied or removed by the electronic circuit, while mobile Ag ions move within the switching medium. The half-reactions alone do not fix filament growth direction: ion supply, electrode kinetics, electron availability and local geometry matter. A VCM alternative invokes redistribution of oxygen-related defects and valence rather than necessarily metallic Ag deposition. Spatially resolved elemental/chemical mapping and in-situ imaging or local electrical measurements, under controlled polarity and history, can discriminate candidate mechanisms; each measurement has resolution and interpretation limits.

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E08

Program separately to distinguishable states, remove programming bias, wait for specified times at controlled temperatures, and read with a verified non-disturbing pulse. Compare state distributions and error probability across devices and cycles. A loop can reflect a slowly relaxing internal state; a zero crossing does not quantify how long it persists. Displacement current during a finite-rate sweep, an electrochemical internal electromotive force, and instrument offset are possible causes of a nonzero measured current at zero terminal voltage; controls are needed to distinguish them.

E09

Convert thickness: $L = 10^{-6}$ cm. Conservation gives $dN/dt = (s - r)/L = 6 \times 10^{18}\,\mathrm{cm}^{-3}\mathrm{s}^{-1}$. The concentration increase to the stated threshold is $3 \times 10^{18}\,\mathrm{cm}^{-3}$; at constant flux, time $= 3/6 = 0.5$ s. This average does not specify the spatial profile, phase, or conducting geometry and cannot establish a local metallic filament.

Redox mechanisms: Worked solutions (continued)

E10

With $v = dV/dt = 1$ V/s, $W = (1/v)\int_0^{V_{\mathrm{end}}} V I(V)\,dV$. For $I = aV$, $W = aV_{\mathrm{end}}^3/(3v)$. Substituting the given values gives SET $W = 1/3\,\mu$J and RESET $W = 4(0.5)^3/3 = 1/6\,\mu$J. The comparison is for the stated voltage intervals, not a general rule. Peak temperature additionally depends on the time profile and heat flow, so energy alone does not determine which peak is higher.

E11

At steady state, $dJ/dx = 0$ and $J = -D dc/dx$, so the concentration is linear: $c(x) = (8 - 6x/L)10^{18}\,\mathrm{cm}^{-3}$. Since $L = 10^{-6}$ cm, the gradient is $-6 \times 10^{24}\,\mathrm{cm}^{-4}$ and $J = +6 \times 10^{10}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}$, toward increasing x. Solving $8 - 6x/L \geq 5$ gives $0 \leq x \leq L/2 = 5$ nm, a 5 nm interval. The calculation alone does not specify a three-dimensional conducting path or establish a connected filament.

E12

Neutrality gives $2[V_O^{\bullet\bullet}] = n + N_A^-$. Thus $n = 2(3 \times 10^{18}) - 2 \times 10^{18} = 4 \times 10^{18}\,\mathrm{cm}^{-3}$. Omitting acceptors incorrectly yields $6 \times 10^{18}\,\mathrm{cm}^{-3}$. Local neutrality can fail within an interface space-charge or depletion region.

E13

At $t = 2$ s the assigned high-concentration interval is $[0, 4]$ nm in A and $[8, 10]$ nm in B; because $8 > 5$, both intervals exceed the threshold and have lengths 4 and 2 nm. Convert 2 and 1 nm/s to 2×10^{-7} and 10^{-7} cm/s. Conservation across the sharp front gives $s = (c_{\mathrm{high}} - c_{\mathrm{low}})v$, so the respective influxes are $(6 \times 10^{18})(2 \times 10^{-7}) = 1.2 \times 10^{12}$ and $(6 \times 10^{18})(10^{-7}) = 6 \times 10^{11}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}$. Under the assigned anchored geometry and redistribution, shrinking by 2 nm leaves A's interval $[0, 2]$ nm and no high-concentration interval in B. The outward flux magnitude is $(6 \times 10^{18})(10^{-7}) = 6 \times 10^{11}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}$ while the front moves. Net flux alone does not uniquely determine the shape, redistribution or front direction, and the assigned speeds cannot be inferred from terminal polarity. This idealized calculation is not a direct reconstruction of a measured filament.

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E14

$L = 10^{-6}$ cm, and conservation gives $d\bar{c}/dt = [J(0) - J(L)]/L = 10^{18}\,\mathrm{cm^{-3}s^{-1}}$. After 2 s the average rises by $2 \times 10^{18}\,\mathrm{cm^{-3}}$. With both boundary fluxes zero, the average is constant in the absence of bulk sources, though internal redistribution can change the local profile. Environment exchange changes the oxide inventory by material transfer; grain-boundary storage and reactive interlayers exchange with a finite reservoir whose own inventory must be tracked; an ideal blocking interface sets the normal flux to zero. Real leakage at electrode edges or exchange with a substrate can invalidate the ideal blocking assumption. One possible $+x$-oriented boundary law at $x = 0$ is $J(0) = k[c_{\mathrm{eq}} - c(0)]$ with k in cm/s; it is a model, not a universal electrode property. Solving $c(x, t)$ additionally needs the transport law, initial condition, the other boundary condition and all relevant coefficients or coupled reservoir kinetics.

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